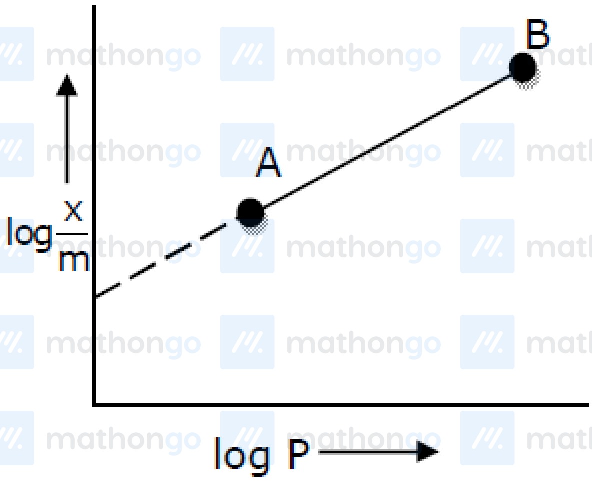
Q1: 24 Feb (Shift 1) - Single Correct

In Freundlich adsorption isotherm, slope of AB line is



(1) $\frac{1}{n}$ with $\left(\frac{1}{n}=0 \text{ to } 1\right)$

(2) $\log \frac{1}{n}$ with $(n<1)$

(3) $\log \mathrm{n}$ with $(\mathrm{n}>1)$

(4) n with $(\mathrm{n}, 0.1 \text{ to } 0.5)$

Q2: 24 Feb (Shift 2) - Single Correct

Most suitable salt which can be used for efficient clotting of blood will be :

(1) $Mg(HCO_3)_2$

(2) $FeSO_4$

(3) $NaHCO_3$

(4) $FeCl_3$

Q3: 25 Feb (Shift 1) - Single Correct

In Freundlich adsorption isotherm at moderate pressure, the extent of adsorption $\left(\frac{x}{m}\right)$ is directly proportional to P^x. The value of X is:

(1) ∞

(2) 1

(3) zero

(4) $\frac{1}{n}$

Q4: 25 Feb (Shift 2) - Single Correct

Which one of the following statements is FALSE for hydrophilic sols ?

(1) These sols are reversible in nature

(2) The sols cannot be easily coagulated

(3) They do not require electrolytes for stability.

(4) Their viscosity is of the order of that of H_2O

Q5: 26 Feb (Shift 2) - Single Correct

The nature of charge on resulting colloidal particles when $FeCl_3$ is added to excess of hot water is:

(1) positive

(2) neutral

(3) sometimes positive and sometimes negative

(4) negative

Answer Key

Q1 (1) **Q2** (4) **Q3** (4) **Q4** (4)

Q5 (1)

Q1 (1)

Freundlich adsorption isotherm is :

$\frac{x}{m} = kp^{1/n}$

$x =$ mass of adsorbate

$\mathrm{m} =$ mass of adsorbent

$\mathrm{P} =$ eq. pressure

$\mathrm{k_1 n} = \frac{1}{\mathrm{n}} \log \mathrm{p} + \log \mathrm{k}$

$y = mx + c$

compairing

$m = \frac{1}{n} = \text{slope} \left[\frac{1}{n} = 0 \text{ to } 1\right]$

$n > 1$

Q2 (4)

Blood is a negative sol. According to hardy-Schulz's rule, the cation with high charge has high coagulation power. Hence, $FeCl_3$ can be used for clotting blood.

Q3 (4)

$\frac{x}{m} = p^x$

the formula is $\frac{x}{m} = p^{1/n}$

Hence $x = \frac{1}{n}$

The value of 'n' is any natural number.

Q4 (4)

Fact base

Q5 (1)

If $FeCl_3$ is added to excess of hot water, a positively charged sol of hydrated ferric oxide is formed due to adsorption of Fe^{3+} ions.